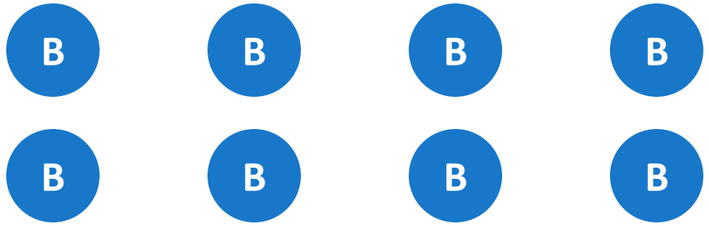
Same color or mixed colors?

Each ball is a sample. Its color is its class. A node is a group of samples.

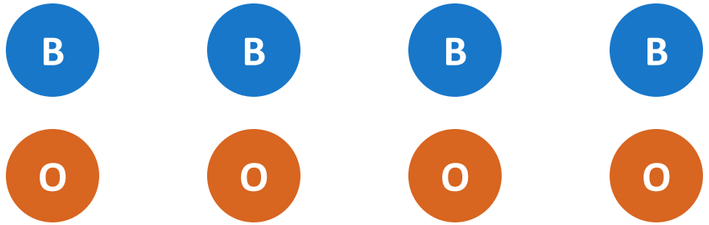
Homogeneous = one class



Pure: every ball is blue.

The next ball must be blue.

Heterogeneous = mixed classes



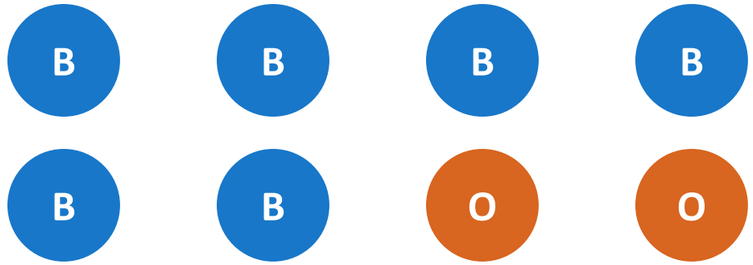
Impure: blue and orange are both present.

The next ball could be either color.

A decision tree looks for feature questions that create purer child nodes.

Probability: count a color, then divide

Probability tells us how likely a color is when we randomly draw one ball.



6 blue + 2 orange = 8 balls

P(color) = count of that color ÷ total

P(B) = 6/8 = 0.75 = 75%

P(O) = 2/8 = 0.25 = 25%

0.75 + 0.25 = 1

A probability of 1 means certain. A probability of 0 means impossible in this group.

Information: surprise from one outcome

An unlikely outcome gives more information when we observe it. A certain outcome gives none.

Information of outcome i: $I(i) = -\log_2(p_i)$

Certain outcome
$p = 1$

$I = -\log_2 1 = 0$ bits

One chance in two
$p = 1/2$

$I = -\log_2(1/2) = 1$ bit

One chance in four
$p = 1/4$

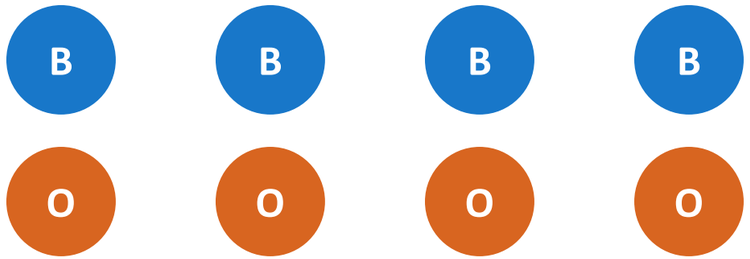
$I = -\log_2(1/4) = 2$ bits

In our 6-blue, 2-orange group, drawing orange gives 2 bits of information.

Base-2 logarithms measure information in bits. Each halving of probability adds 1 bit.

Entropy: average information before a draw

Entropy measures uncertainty about the class of the next sample. It averages the surprise of each outcome.



P(B) = P(O) = 4/8 = 0.5

Either color gives 1 bit of information.

$$H = \Sigma\, p_i\, I(i) = -\Sigma\, p_i \log_2(p_i)$$

Blue contribution: 0.5 × 1 = 0.5

Orange contribution: 0.5 × 1 = 0.5

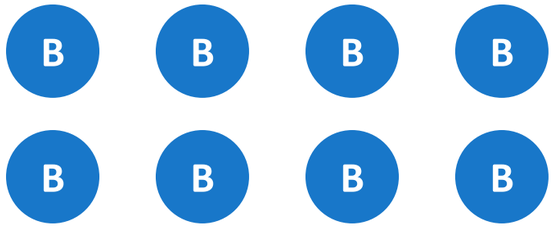
H = 0.5 + 0.5 = 1 bit

Information is surprise from one outcome. Entropy is the average surprise of the group.

Entropy is highest when the two classes are balanced

For two classes, entropy ranges from 0 bits (pure) to 1 bit (a 50–50 mix).

8 blue, 0 orange

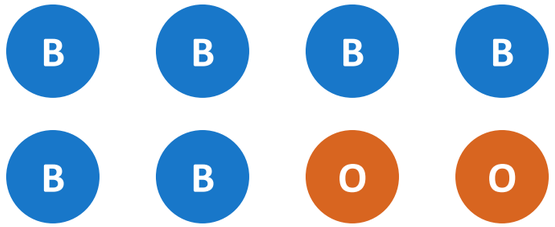


P(B) = 1, P(O) = 0

H = 0 bits

No uncertainty

6 blue, 2 orange

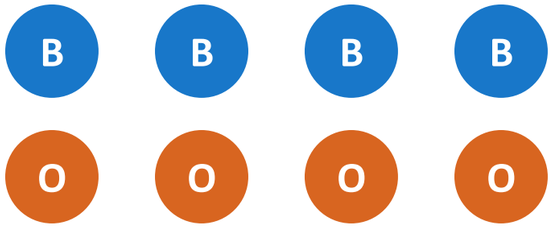


P(B) = 0.75, P(O) = 0.25

H ≈ 0.811 bits

Some uncertainty

4 blue, 4 orange



P(B) = 0.5, P(O) = 0.5

H = 1 bit

Maximum uncertainty

For 6/2: $H = -[0.75 \log_2 0.75 + 0.25 \log_2 0.25] \approx 0.811$. Use $0 \log_2 0 = 0$.

Information gain: uncertainty removed by a split

A feature question is useful when its child nodes have lower weighted entropy than the parent.

Parent: 4 blue + 4 orange



Parent entropy = 1 bit

Feature question: weight ≤ 5 g?

Yes: four blue balls



H(left) = 0

No: four orange balls



H(right) = 0

Weighted child entropy
= (4/8) × 0 + (4/8) × 0
= 0 bits

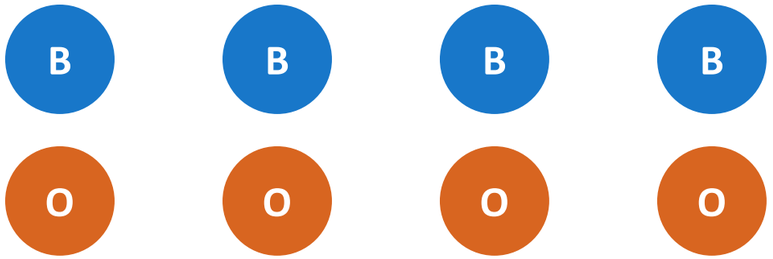
Information gain = 1 – 0
= 1 bit

Information gain = parent entropy – weighted child entropy. Larger gain is better.

Gini impurity: how mixed are the classes?

Gini equals the chance of different colors in two independent draws with replacement.

With replacement: draw a ball, put it back, then draw again.



$$G = 1 - \Sigma\, p_i^2$$

Chance of matching colors
$= P(B)^2 + P(O)^2$
$= 0.5^2 + 0.5^2 = 0.5$

Pure node: $G = 1 - (1^2 + 0^2) = 0$

Chance of different colors
$G = 1 - 0.5 = 0.5$

For two classes: $0 \le G \le 0.5$. Lower Gini means a purer node.

Entropy and Gini: two ways to score a split

A tree usually uses one criterion. It compares candidate splits and chooses the largest impurity reduction.

Entropy

Meaning: average surprise
Pure node: 0
50–50 binary node: 1 bit
Split score: information gain

A pure split of a 4/4 parent:
Information gain = 1; Gini decrease = 0.5

Gini

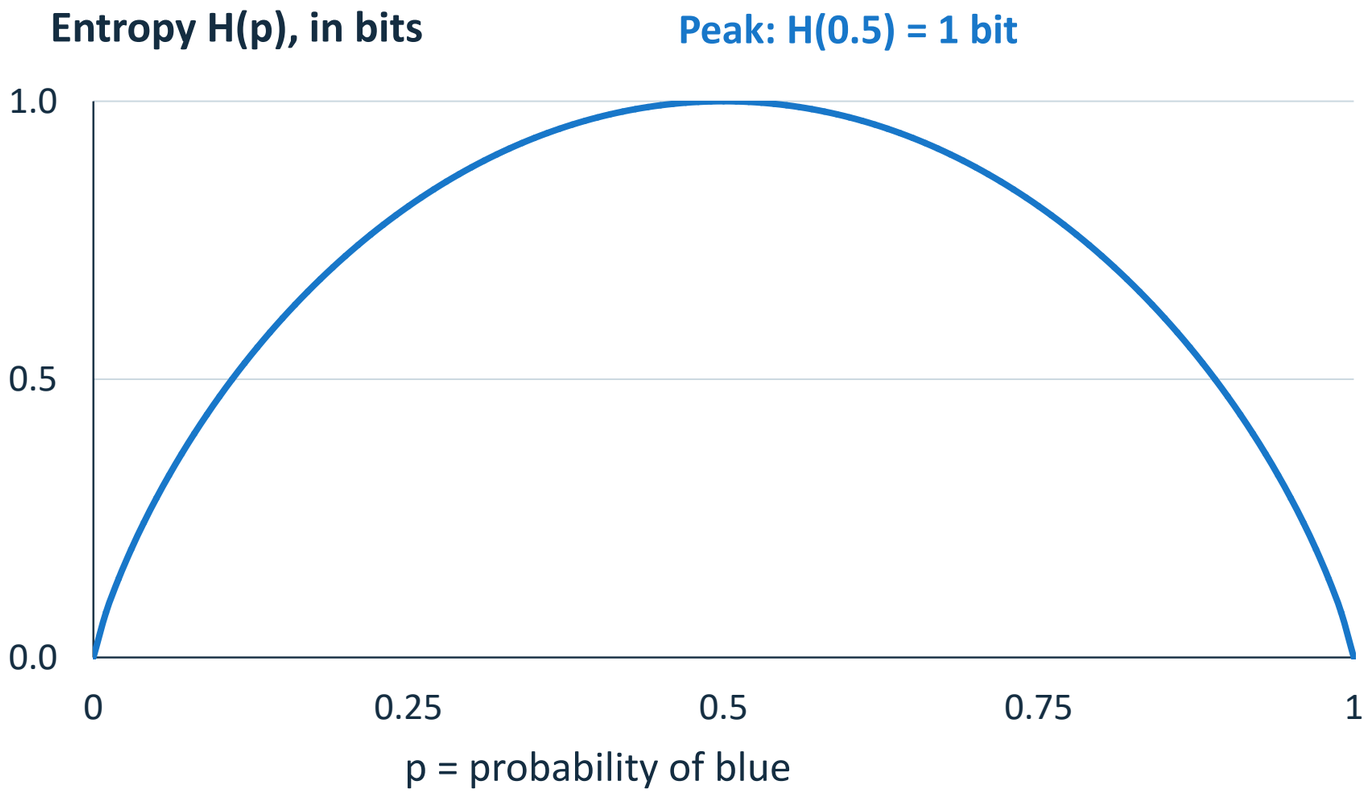
Meaning: probability of different classes
Pure node: 0
50–50 binary node: 0.5
Split score: Gini decrease

A split into two 2/2 mixed children:
Information gain = 0; Gini decrease = 0

Compare splits using the same criterion. Pure training nodes do not guarantee good test performance.

Shannon entropy: uncertainty in one number

Claude Shannon introduced this measure in 1948. For two colors, a 50–50 mix has the most uncertainty.





p = 0 H = 0
Certain: orange



p = 0.5 H = 1 bit
Most uncertain



p = 1 H = 0
Certain: blue

$H(p) = -p \log_2 p - (1 - p) \log_2(1 - p)$. At the ends, use $0 \log_2 0 = 0$.

Mutual information: how much does a clue tell us?

Let Y be the ball's color. Let X be the answer to a feature question. Start with four blue and four orange.

I(X; Y) = uncertainty before − uncertainty after = H(Y) − H(Y | X)

Useful question: weight ≤ 5 g?

Yes



No



After: both groups are pure
H(Y | X) = (4/8)×0 + (4/8)×0 = 0
I(X; Y) = 1 − 0 = 1 bit

Useless question: box side?

Left side



Right side

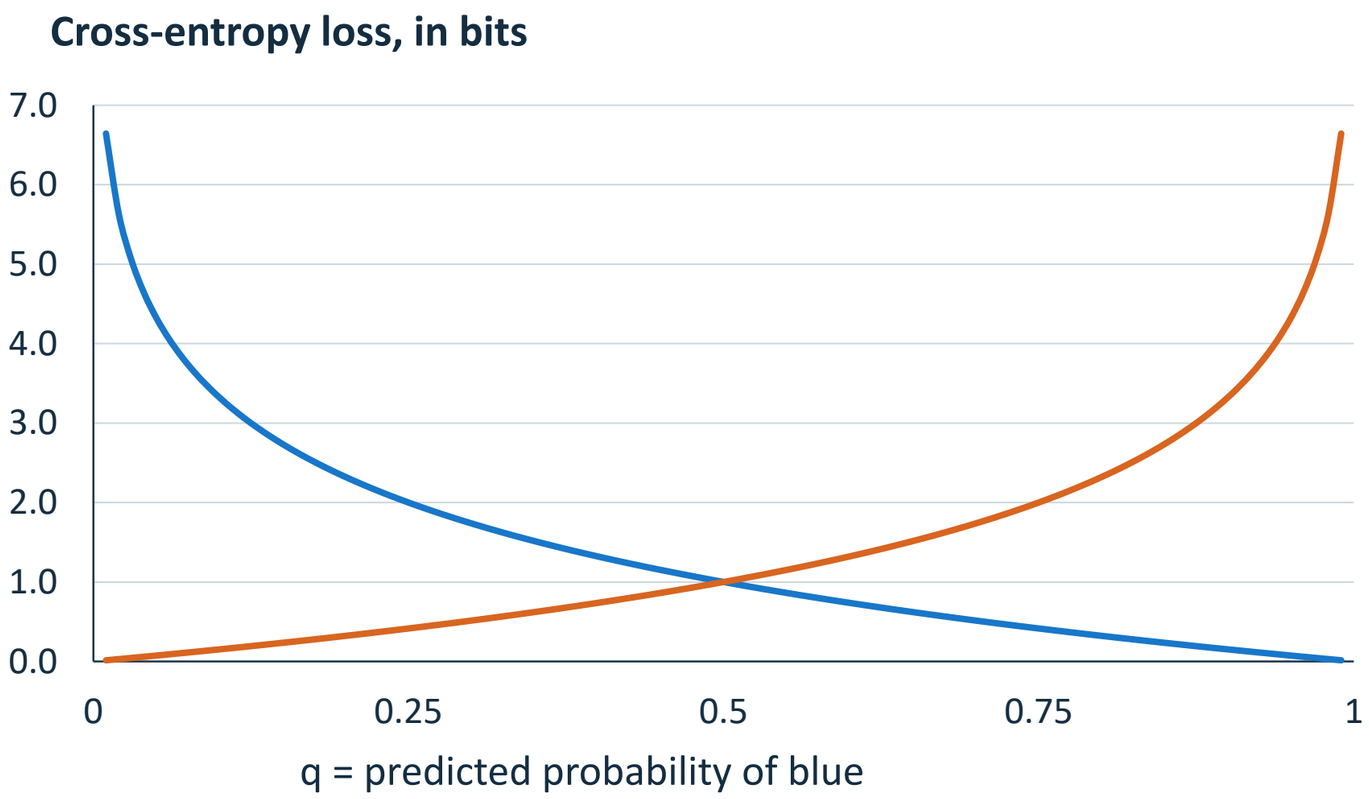


After: both groups are 50–50
H(Y | X) = (4/8)×1 + (4/8)×1 = 1
I(X; Y) = 1 − 1 = 0 bits

For a tree split, information gain equals I(X; Y). More gain means less uncertainty after the split.

Binary cross-entropy: the cost of a prediction

q is the model's predicted probability of blue. The true label y is 1 for blue and 0 for orange.



True blue: $L = -\log_2 q$

True orange: $L = -\log_2(1 - q)$

At q = 0.5: loss = 1 bit
The model gives equal chances.

Confident and correct: loss near 0
Confident and wrong: very large loss
Training aims to minimize this loss.

Entropy peaks at a 50–50 mix. Trees maximize information gain. Prediction training minimizes cross-entropy.

Concept check: three multiple-choice questions

Choose one answer per question. Each click reveals one answer and its explanation.

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1. Which group is homogeneous?

A. Eight blue balls B. Four blue and four orange
C. Six blue and two orange D. One blue and seven orange
Answer: A. A homogeneous group contains only one class.

2. Which observed outcome carries the most information?

A. 100% chance B. 25% chance C. 50% chance D. 75% chance

Answer: B. The least likely outcome is most surprising: $-\log_2(1/4) = 2$ bits.

3. What does entropy measure?

A. Number of balls B. Largest class probability C. Average surprise D. Tree depth

Answer: C. Entropy averages each class's information, weighted by its probability.

Calculation check: probabilities, Gini, and gain

Use $G = 1 - \Sigma p_i^2$ and information gain = parent H − weighted child H.

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4. A node has 6 blue and 2 orange balls. What is its Gini impurity?

A. 0.25 B. 0.375 C. 0.5 D. 0.75

Answer: B. $G = 1 - (0.75^2 + 0.25^2) = 1 - 0.625 = 0.375$.

5. A node has 4 blue and 4 orange balls. What is its entropy?

A. 0 bits B. 0.5 bits C. 1 bit D. 2 bits

Answer: C. $H = -[0.5(-1) + 0.5(-1)] = 1$ bit.

6. Parent H = 1 bit. Weighted child H = 0.4 bits. What is the gain?

A. 0.4 bits B. 1.4 bits C. 1 bit D. 0.6 bits

Answer: D. Information gain = 1 − 0.4 = 0.6 bits.